

Wireshark

- **User Interface:** Features a rich **Graphical User Interface (GUI)** that displays traffic in three main panes: a packet list, packet details, and a hex dump.
- **Analysis Capabilities:** Includes advanced features like **color-coding** to highlight different traffic types and the ability to **"Follow TCP Streams"** to reconstruct entire conversations between hosts.
- **Software Requirements:** Open-source and available for Windows, macOS, and Linux. It is more resource-heavy than tcpdump due to its graphical nature.

Similarities

- **Library Dependency:** Both tools use the pcap (packet capture) library (libpcap or WinPcap/Npcap) to read data from network interfaces.
- **File Compatibility:** Both support the .pcap or .pcapng file format. Traffic captured by tcpdump can be opened in Wireshark for visual analysis.
- **Passive Nature:** Both are passive sniffers; they only listen to and record the traffic that reaches the monitored interface and do not actively block or stop attacks.

tcpdump

- **User Interface:** Operated entirely through a **Command-Line Interface (CLI)**; it displays packet headers and data directly in the terminal.
- **Usage Scenarios:** Ideal for **remote servers** or "headless" systems where a GUI is unavailable. Analysts often use it to capture traffic quickly and save it to a file for later analysis.
- **Software Requirements:** Open-source and pre-installed on many Unix-like systems, including macOS and most Linux distributions.